Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/28/2007
PI Name:	Jacobs, Jeffrey Ph.D.		
Project Title:	An Experimental Investigation of Incompressible Richtmyer-Meshkov Instability		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/18/2002	**End Date:**	11/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Niederhaus, Charles (NASA Glenn Research Center)		
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Performance Goal Text:			
	Richtmyer-Meshkov instability occurs when a planar interface separating two fluids of different densities is impulsively accelerated in the direction of its normal. The focus of this research project is to observe and quantify the nonlinear development of this instability from a simple well characterized initial state into a turbulent flow, and to determine whether a self-preserving final turbulent state is achieved. The experiments utilize a novel experimental apparatus which circumvents many of the experimental difficulties that have previously limited the study of Richtmyer-Meshkov instability. In this system, the instability is generated by impulsively accelerating a rectangular box containing two liquids. The effects of gravity are minimized by keeping the fluids in free-fall. Previous NASA funding supported the construction of a laboratory apparatus in which the impulsive acceleration is imparted by bouncing the fluid box off of a fixed spring. The current project will extend the previous study and focus on the turbulent regime to determine self-similarity and to obtain the turbulent growth rate. The extended duration of microgravity offered by the 2.2 Second		

Task Description:	Drop Tower at GRC and the KC-135 Reduced Gravity Aircraft will be utilized to study the turbulent regime. An adaptation of the existing laboratory apparatus for use in the 2.2 Second Drop Tower was initiated under previous NASA funding. Experiments will therefore initially be carried out in this facility using the new apparatus and later progress to the Reduced Gravity Aircraft. The adaptation of the experiments to the KC-135 will be aided by experiments at the University of Arizona that will focus on developing techniques for initial interface formation using paramagnetic fluids. These methods will also prove useful for a possible future space experiment. The objective of the funded research is to carry out ground-based experiments to study the nonlinear development and transition to turbulence of Richtmyer-Meshkov instability with the goal of quantifying the development of the resulting turbulent flow and to determine if a self-similar state is achieved. The long term objective of this project is to develop methods and procedures for a possible future flight experiment in order utilize the low gravity environment of space for the creation of an experiment with virtually unlimited duration to study the fully turbulent Richtmyer-Meshkov instability. Richtmyer-Meshkov instability is one of the most fundamental of fluid instabilities and is of importance in fields ranging from astrophysics to materials processing. For example, Richtmyer-Meshkov instability is of critical importance to inertial confinement fusion. The degree of compression capable in laser fusion experiments is ultimately limited by Richtmyer-Meshkov instability and another similar instability called Rayleigh-Taylor instability. Thus, these instabilities represent the single most significant barrier to attaining positive-net-yield fusion reactions in laser fusion facilities. Richtmyer-Meshkov instability is also of importance to high speed combustion applications, where, for example, it can greatly enhance mixing and thus significantly alter the burning rate. Knowledge gained from this study should aid in the design of laser fusion targets and supersonic combustors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Richtmyer-Meshkov instability is one of the most fundamental of fluid instabilities and is of importance in fields ranging from astrophysics to materials processing. For example, Richtmyer-Meshkov instability is of critical importance to inertial confinement fusion. The degree of compression capable in laser fusion experiments is ultimately limited by Richtmyer-Meshkov instability and another similar instability called Rayleigh-Taylor instability. Thus, these instabilities represent the single most significant barrier to attaining positive-net-yield fusion reactions in laser fusion facilities.
Task Progress:	Work the past year has been focused on the development of our capability to produce stable interfaces in microgravity using magnetic fluids. Earlier work in this effort has concentrated on evaluating electromagnet configurations for this purpose, modeling them using the finite element package ANSYS. Using the results of this analysis it was decided to implement a dipole electromagnet configuration and we have recently installed a large (10” pole diameter) electromagnet with power supply. More recent work has been focused on using ANSYS to determine the optimum pole face geometry to carry out the dipole experiments. In this study field distributions were computed to determine the best pole contour for providing a large field gradient while also providing nearly horizontal field contours. The results of this research have indicated that a cosine pole face satisfies our requirements and this shape has subsequently been machined onto the poles of our electromagnet. Preliminary experiments utilizing this configuration have demonstrated its ability to stabilize a heaver magnetic fluid on top of a lighter magnetic one, and have confirmed our calculations predicting that this choice of pole geometry does generate a flat interface. As a result, we have recently begun an experimental investigation of the magnetically stabilized Rayleigh-Taylor instability utilizing this system. In these experiments the liquids, a miscible combination of paramagnetic salt solution and one of three nonmagnetic solutions, are contained in a tank placed between the poles of our electromagnet. First the nonmagnetic lighter liquid is placed in the bottom half of the tank. Then, after the magnetic field is turned on, the heavier magnetic liquid is placed on top. The magnetic field is capable of suspending the heavy paramagnetic fluid over the lighter non-magnetic fluid utilizing the gradient field principle. Rayleigh-Taylor instability is then initiated by rapidly removing power to the electromagnet which results in the heavy fluid falling under the influence of gravity. The resulting instability is visualized using backlit photography. Experiments initiated with an initially flat interface develop into a random surface pattern with a dominant length scale well approximated by the fastest growing wavelength given by viscous linear stability theory. Measurements of the mean mixing zone width are observed to follow a gt2 dependence with a constant of proportionality that falls within the range found in previous experimental studies. This experimental program has continued this past year with the implementation of PLIF flow visualization. To accomplish this Rhodamine 6G dye is mixed with the lower fluid and an Argon Ion laser is used to form a thin light sheet which is passed through the top of the tank, illuminating a thin cross section of the two fluid mixture. The fluorescent signal is then recorded with a digital CCD camera. The implementation of PLIF is an improvement because it allows the observation of a thin cross section of the flow.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Abstracts for Journals and Proceedings	Olson D, Roberts M, Romero-Colio P, Jacobs J. "Experiments on the Rayleigh-Taylor and Richtmyer-Meshkov instabilities." Presented at the 10th International Workshop on the Physics of Compressible Turbulent Mixing, Paris, France, July 17-21, 2006. 10th International Workshop on the Physics of Compressible Turbulent Mixing, conference proceeding, July 2006. , Jul-2006
Abstracts for Journals and Proceedings	Gohardani O, Oemke R, Jacobs J. "PLIF flow visualization of magnetically stabilized Rayleigh-Taylor instability." Presented at the 59th Meeting of the APS/DFD, Tampa, Florida, November 19-21, 2006. Bull Am Phys Soc. 2006 Nov; 51(9): 248P. , Nov-2006
Articles in Peer-reviewed Journals	Chapman PR, Jacobs JW. "Experiments on the three-dimensional incompressible Richtmyer-Meshkov instability." Phys Fluids. 2006 Jul;18(7):074101. http://dx.doi.org/10.1063/1.2214647 , Jul-2006